

A Split-Merge Clustering Algorithm based on Unimodality Testing

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**ΤΜΗΜΑ ΜΗΧ. Η/Υ & ΠΛΗΡΟΦΟΡΙΚΗΣ
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Giannis Fillis

Περίληψη

Στην παρούσα διπλωματική προτείνουμε μία νέα μέθοδο ομαδοποίησης δεδομένων η οποία βασίζεται στην ήδη υπάρχουσα μέθοδο UniForCE. Ο UniForce είναι ένας αλγόριθμος ομαδοποίησης ο οποίος ανακαλύπτει τοπικά μονοτροπικές συστάδες ακολουθώντας μία bottom-up προσέγγιση: Ξεκινά με μία αρχική υπερ-συσταδοποίηση του συνόλου δεδομένων σε πολλές μικρές υπο-συστάδες και έπειτα τις συγχωνεύει σταδιακά, εφόσον η ένωσή τους διατηρεί την μονοτροπικότητα. Αυτή η διαδικασία δημιουργεί ένα δάσος με δέντρα, όπου κάθε δέντρο αποτελείται από πολλές υπο-συστάδες και αντιπροσωπεύει μία τελική συστάδα.

Ένας περιορισμός που έχει είναι ότι θεωρεί πως οι αρχικές υπο-συστάδες που δημιουργεί είναι ομοιογενής. Εμείς, προτείνουμε μία βελτιωμένη έκδοση, όπου αμέσως μετά την δημιουργία των υπο-συστάδων εφαρμόζουμε τον στατιστικό έλεγχο μονοτροπικότητας mud-rod και αν κάποια από αυτές βρεθεί πολυτροπική, που σημαίνει ότι δεν είναι ομοιογενείς, διασπάται περαιτέρω. Έτσι εξασφαλίζουμε την ομοιογένεια των υπο-συστάδων.

Πραγματοποιήσαμε πολλά πειράματα, συγκρίνοντας την μέθοδο μας με την αρχική τόσο σε πραγματικά όσο και σε συνθετικά σύνολα δεδομένων. Τα αποτελέσματα έδειξαν πως ο αλγόριθμος μας είχε καλύτερη ή και συγκρίσιμη επίδοση στα περισσότερα σύνολα. Επίσης, αυτό που συμπεράναμε είναι πως ο αλγόριθμος χαλαρώνει την εξάρτηση ως προς την υπερ-παράμετρο K , που είναι το πλήθος των αρχικών υπο-ομάδων, καθώς με την εφαρμογή του mud-rod μπορεί να προχωρήσει σε περαιτέρω διάσπαση των ομάδων αν χρειαστεί.

Λέξεις Κλειδιά: Ομαδοποίηση, Έλεγχος Μονοτροπικότητας, Τυχαίες Προβολές, Ομαδοποίηση Συστάδων, Τοπικά Μονοτροπική Συστάδα

Abstract

In this thesis we propose a new clustering method which is based on the already existing UniForCE method. UniForce is a clustering method which operates by discovering locally unimodal clusters through a bottom-up approach: it begins by partitioning the dataset into numerous small subclusters and subsequently merges them, provided their union preserves unimodality. This process creates a forest of trees, where each tree comprises multiple subclusters and represents a final cluster.

One of the limitations is that it assumes that the initial subclusters it creates are homogeneous. We propose an improved version where immediately after the creation of the initial subclusters we apply the mud-pod unimodality test and if a cluster is categorized as multimodal, meaning it is not homogeneous, it is further partitioned. This way, we ensure that the subclusters are homogeneous.

We conducted a lot of experiments, comparing our method with the initial UniForCE in both real-world and synthetic datasets. The results indicate that our algorithm consistently achieves comparable or higher performance in most datasets. Furthermore, what we concluded is that our method relaxes the dependency on the hyperparameter K , which is the number of initial subclusters, since with the application of mud-pod it can continue with the further partition of the subclusters.

Keywords: Clustering, Unimodality Test, Random Projections, Cluster Aggregation, Locally Unimodal Cluster

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Chapter 1. Introduction

1.1 Motivation

Clustering is a fundamental unsupervised machine learning problem that aims to uncover the hidden structure and patterns within a dataset. One of the main advantages of this method is that it requires minimal prior knowledge and lacks labeled examples. Because of this, it is inherently an ill-posed problem: there are many different but mathematically “reasonable” objectives to optimize [\[1\],\[2\]](#). To make the problem manageable, clustering algorithms rely on “cluster assumption”, which defines the mathematical or structural characteristics a subset of data must exhibit to be considered a valid cluster. However, defining what constitutes a meaningful cluster becomes exceedingly difficult when dealing with real-world data.

Determining the optimal number of clusters, denoted as k , during the optimization procedure remains one of the most challenging problems in the field, particularly in high-dimensional spaces. Most traditional methods either require k to be provided as a user input or rely on heuristic hyperparameters that shift the difficulty of the problem rather than solving it. Furthermore, there is a strong interdependency between the cluster assumption and the estimation of k : if an algorithm assumes an inadequate cluster shape, the estimation of the number of clusters is highly likely to be incorrect, resulting in a final model which is less informative.

Standard clustering models frequently fall short when dealing with real-world data consisting of irregular, non-convex shapes, primarily because they rely on geometric assumptions like sphericity or Gaussianity. As a more versatile alternative, density-shape assumptions based on unimodality have gained traction. By evaluating whether data points converge towards a single central mode or form distinct high-density regions, unimodality offers a robust framework for identifying complex cluster structures without constraining their overall geometry.

Nevertheless, extending unimodality testing from one-dimensional data, where tools like Hartigan’s dip test are mathematically mature, to multivariate spaces introduces profound mathematical challenges. Existing multivariate approaches, such as the dip-dist criterion, function as ad-hoc heuristics within the original feature space lacking the theoretical rigor needed for consistent performance. This gap highlights a clear necessity for an advanced clustering methodology: one that can accommodate complex, arbitrary cluster shapes and determine the optimal cluster count by embedding a mathematically robust, high-dimensional unimodality test directly into its core decision making process.

1.2 Problem Statement

The success of bottom-up clustering methods that utilize local density-shape properties, such as the UniForCE framework, depends on the structural integrity and homogeneity of their initial clusters created. UniForCE operates by first overclustering the dataset into a set of subclusters, which are theoretically assumed to reside within strictly homogeneous and convex subregions of the data space. This initial partitioning phase relies on distance-based partitioning algorithms like the global k-means++. However, when applied to high-dimensional settings or datasets characterized by complex cluster boundaries and significant noise, the performance of standard partitioning algorithms can heavily deteriorate. Consequently, the resulting initial subclusters may inadvertently encompass multiple internal density modes or blend overlapping regions from distinct underlying clusters.

The core vulnerability of this approach lies in the absence of a statistical verification of each individual subcluster’s internal modality prior to the graph aggregation phase. While the subsequent stages of the framework evaluate the unimodality of combined subcluster pairs to detect density gaps, they proceed under the implicit assumption that the subclusters created at the previous step are already unimodal. If this assumption is violated at the very beginning, an inadequate representation of the local data topology occurs. This structural flaw significantly increases the probability that the subsequent tree formations are distorted, ultimately leading to a wrong estimation of the final number of clusters k .

Therefore, the specific problem addressed in this thesis is the lack of a mathematically grounded validation mechanism of enforcing rigorous multidimensional unimodality

constraints on the initial overclustered components. To resolve this limitation, a theoretically consistent multivariate unimodality test must be integrated directly into the overclustering pipeline as a strict validation filter. The fundamental challenge involves effectively leveraging linear random projections and Mahalanobis distances via the mud-pod test to mathematically verify and refine these initial subclusters. This structural enhancement ensures that the bottom-up forest, is built upon statistically guaranteed and strictly unimodal foundational units, thereby preserving the accuracy of both data partitioning and automated cluster estimation.

1.3 Thesis Contribution

This thesis bridges the gap between flexible, bottom-up cluster aggregation and rigorous multivariate statistical validation. To achieve this, we propose an improved version of the Unimodality Forest for Clustering Estimation method (UniForCE) where we integrate the mud-pod test. By embedding multivariate statistical unimodality constraints directly into a bottom-up graph-based clustering architecture, this novel pipeline successfully leverages the shape-flexibility of local density representations while maintaining high mathematical rigor throughout the partitioning process.

A key aspect of our contribution involves enhancing the foundational overclustering phase of the UniForCE algorithm. Instead of relying blindly on the geometric distance metrics of standard partitioning algorithms like global k-means++, we introduce the mud-pod test as an intermediate validation filter to statistically verify and refine the internal unimodality of each initial subcluster before any graph-merging decisions take place. By utilizing linear random projections and Mahalanobis distances at this early stage, we effectively mitigate partitioning errors caused by complex topologies and noise. This structural improvement prevents the propagation of geometric defects during the forest creation, which significantly limits false positive cluster merges and increases the reliability of the automated cluster count estimation k .

We conducted an extensive evaluation of the proposed algorithm across diverse synthetic and complex real-world benchmark datasets. The experimental results suggest that in many cases it outperforms the initial UniForCE framework providing better cluster estimation and accuracy. In addition to shape flexibility, our proposed architecture introduces significant robustness regarding the choice of the initial overclustering parameter K . By leveraging the mud-pod test as an automated quality

control filter, our method actively detects and refines these multi-modal subclusters. This statistical intervention relaxes the dependency on a perfectly tuned initial K , ensuring that even a suboptimal user-defined setting is dynamically corrected to preserve the accuracy of the subsequent forest and the automated estimation of k .

1.4 Thesis Structure

This thesis is structured in the following chapters:

In chapter 1, we begin with a brief introduction into the problem of clustering and its challenges. We present the motivation of this work and also the problem we are solving.

In chapter 2, we delve deeper into the foundational background we need. We present in detail the concept of clustering, the distance metrics used and the objective functions. We also mention the cluster assumptions which are commonly used. We analyze the unimodality in both one-dimensional and multidimensional data and finally the fundamentals of graph theory.

In chapter 3, we analyze in detail the UniForCE algorithm and the steps it follows to perform the clustering. We also analyze the mud-pod test and how it utilizes the Mahalanobis distances and the random projections to detect multimodality in a dataset.

In chapter 4, we present our proposed framework and how we integrated the mud-pod test immediately after the overclustering step of the UniForCE algorithm.

In chapter 5, we present a series of experiments that we conducted. Firstly, we show the setup of the experiments and then we analyze the experiments on both real-world and synthetic datasets. We also performed some experiments on the datasets to find the optimal value of the hyperparameter K and compared the performance of the two algorithms on their respective optimal value of K . Lastly, we discuss the limitations of our approach that we found in these experiments.

In chapter 6, we summarize the key findings and contributions of this thesis.

Chapter 2. Background

2.1 Data Clustering Fundamentals

2.1.1 The Concept of Clustering

As mentioned before, clustering is a fundamental unsupervised machine learning problem, aiming to uncover the hidden structure and patterns within unlabeled datasets. Formally, given a dataset $X = \{x_1, x_2, \dots, x_N\}$ consisting of N data points in a d -dimensional real space $\mathbb{R}^d$, clustering is defined as the process of partitioning this set of objects into groups, known as clusters. A valid partition $C = \{C_1, C_2, \dots, C_k\}$ divides the dataset into k distinct, pairwise disjoint subsets whose union represents the entire dataset X . The primary objective is to group the data such that instances within the same cluster share common characteristics (high intra-cluster similarity) while differing significantly from instances in other groups (low inter-cluster similarity).

Despite its straightforward definition, the absence of labeled examples makes clustering a fundamentally challenging problem. To make the problem easier, algorithms must rely on a mathematical precondition known as the cluster assumption, which dictates the expected characteristics that a subset of data should exhibit to qualify as a cluster.

2.1.2 Distance Metrics and Objective Functions

At the core of any clustering algorithm is the definition of similarity, which is mathematically expressed through distance metrics. The choice of distance metric heavily influences the shape of the clusters the algorithm can identify. The most used metric is the Euclidean distance, which measures the straight-line distance between two points x and y :

$$D(x, y) = \sqrt{(x - y)^T (x - y)}$$

Algorithms like k-means utilize the Euclidean distance to minimize the within-cluster variance, commonly referred to as the Sum of Squared Errors (SSE):

$$SSE = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

Where μ_i represents the centroid of cluster C_i .

However, the Euclidean distance assumes that data is spherically distributed and equally scaled across all dimensions. In real world multivariate datasets, features often exhibit different variances and correlations. To address this, the Mahalanobis distance can be utilized, which considers the covariance structure across all dimensions. Given a distribution with a mean vector μ and a positive semidefinite covariance matrix Σ , the Mahalanobis distance of a point x from the mean is defined as:

$$D(x, \mu) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}$$

The Mahalanobis distance possesses three distinct properties: it is unitless, scale-invariant and accurately captures the elliptical shape of data distributions. This metric forms a critical foundation for advanced multivariate statistical testing, as it provides a more accurate representation of structural distances in complex data spaces.

2.1.3 Classical Cluster Assumptions

Because different algorithms use different metrics and optimization goals, they adopt distinct cluster assumptions.

First, we have the prototype-based assumptions. These models partition data around objects that can be centroids [3], medoids [4] or other exemplars [5], [6], [7], [8]. Due to their reliance on central points and symmetric distance metrics, prototype-based assumptions inevitably lead to convex-shaped clusters.

Next, we have the model-based assumptions. These approaches consider a probabilistic model for each cluster, such as Gaussian [9] or linear models [10]. They assume the dataset is generated by a mixture of underlying probability distributions. For instance, Gaussian mixture modeling will always fit the data with Gaussian components regardless of the actual validity of this assumption.

Lastly, we have the density-based assumptions. These methods depart from strict geometric boundaries and define clusters as continuous regions of high local density separated by lower-density areas [11].

While these classical assumptions provide robust frameworks for specific types of data, they often fail when faced with irregular cluster shapes. The density shape assumptions whether they are implicit, like the convex shapes in prototype-based models or explicit, like the Gaussianity in model-based methods, restrict the algorithms' ability to model complex topologies. This limitation creates the necessity for a shift towards more flexible density-shape assumptions, such as unimodality [12], which allows for arbitrary cluster geometries as long as a single prominent peak of density is maintained.

2.2 Unimodality and Statistical Assumption

2.2.1 The Unimodality Cluster Assumption

As highlighted in the previous sections, when we rely on geometric metrics, we restrict the clustering algorithm's ability to discover complex and real-world structures. To overcome this, we need to shift the focus towards analyzing the density profile of the data. The fundamental principle of this approach is the unimodality cluster assumption.

In statistical terms, a dataset (or a subset of it) is considered to form a single cluster if its underlying probability density function (PDF) is unimodal, meaning it possesses exactly one prominent local maximum (a single peak of density). In contrast, if the density function exhibits multiple peaks (multimodality), it strongly indicates the presence of multiple overlapping or distinct clusters that must be separated. The principal advantage of the unimodality assumption is its flexibility. A unimodal distribution can be spherical, elliptical or entirely asymmetrical, provided that the data points converge towards a single continuous high-density region.

2.2.2 Mathematical Formulation of 1D Unimodality

To mathematically validate unimodality, we must first define it in a one-dimensional (1D) space. Let X be a one-dimensional continuous random variable with a cumulative distribution function (CDF) denoted as $F(x)$ and a corresponding probability density function $f(x)$. The distribution is defined as unimodal with a mode at m if its density function $f(x)$ is monotonically non-decreasing for all $x \leq m$ and monotonically non-increasing for all $x \geq m$.

An alternative definition considers the cumulative distribution function (CDF). A c.d.f. is said to be unimodal if there exist two points, x_l and x_u , such that the function can be split into three parts: a) a convex part $(-\infty, x_l]$, b) a constant part $[x_l, x_u]$ and c) a concave part

(x_u, ∞) . It is possible though for either of the first two parts or the last two parts to be missing.

In [Figure 2.1](#) [\[27\]](#) we illustrate unimodality in some of the most well-known distributions and their respective c.d.f. at the bottom. We can see that the Gaussian has a peak at around 2, the student's t has a steep and tight peak at zero and the Gamma distribution while being skewed with its tail extending to the right also exhibiting a mode at around 4. Lastly, we have the uniform distribution which is considered an extreme case of single mode case.

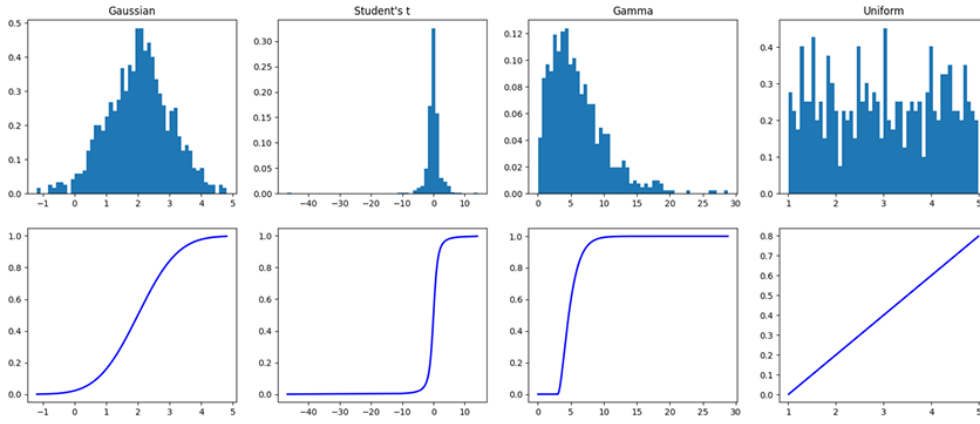


Figure 2.1. Histogram plots of unimodal distributions (top row) and corresponding cdf plots (bottom row).

2.2.3 Hartigan's Dip Test

To objectively determine whether a given sample of data is drawn from a unimodal distribution, a robust statistical hypothesis test is required. The most prominent and widely adopted non-parametric method for this purpose is the dip test, introduced by Hartigan and Hartigan [\[13\]](#). Unlike other approaches that test against specific parametric distributions (e.g. Gaussian), the dip test evaluates the data against the entire class of unimodal distributions and as such it is considered very flexible.

The core concept of the dip test revolves around the empirical cumulative distribution function (e.c.d.f.), denoted as $F_n(x)$, which is constructed from the observed data sample. The dip statistic measures the maximum difference between the empirical distribution F_n and the closest possible perfectly unimodal distribution G . Mathematically, the dip statistic D is defined as:

$$D(F_n) = \min_{G \in U} \max_x |F_n(x) - G(x)|$$

Where U represents the class of all unimodal distribution functions.

Essentially, the dip statistic computes the minimum among the maximum deviations observed between the c.d.f. F and the c.d.f.s from the class of unimodal distributions. An interesting property of dip is that if we take n samples from F , then $\lim_{n \rightarrow \infty} \text{dip}(F_n) = \text{dip}(F)$.

To determine statistical significance, we compare the calculated dip value against a null distribution, typically a uniform distribution, which is empirically considered the worst-case unimodal distribution. We then extract a p-value using techniques like bootstrapping. If the p-value falls below a predefined significance level α , the null hypothesis of unimodality is rejected, confidently indicating that the data is multimodal.

The elegance of Hartigan's dip test lies in its non-parametric nature and mathematical robustness. It provides a definite "yes or no" answer to whether a data subset should be split. However, the original dip test is strictly defined for one-dimensional data. Real-world clustering involves multidimensional spaces ($d > 1$), where measuring convexity and concavity becomes mathematically difficult to deal with. Therefore, the subsequent challenge and the foundation of modern frameworks like mud-pod is developing processes to effectively map multidimensional relationships to 1D representations, where the dip test can be executed.

2.3 Multivariate Unimodality

2.3.1 The Challenge of Multidimensional Modality

While Hartigan's dip-test provides a mathematically optimal framework for assessing unimodality, its application is strictly limited to one-dimensional data. The extension of the concept of unimodality to multivariate spaces ($d > 1$) introduces profound theoretical and computational challenges. The core issue lies in the fundamental definition of the empirical cumulative distribution function (e.c.d.f.). In a 1D space, data points possess a natural mathematical ordering (from smallest to largest), allowing for the clear definition of convexity and concavity required by the dip statistic. In multidimensional spaces, however, this natural ordering usually does not exist. As a result, assessing whether a multivariate density surface contains a single high-density peak requires alternative transformations to map the multidimensional topological relationships into 1D distributions which can be evaluated.

2.3.2 The α -Unimodality

To overcome the limitations of the 1D empirical cumulative distribution function, statisticians had to redefine what constitutes a “mode” in a multidimensional space in $\mathbb{R}^d$. The most robust formulation for this is the concept of α -unimodality, introduced by Olshen and Savage [14].

Formally, without loss of generality, assuming the mode of a distribution is located at the origin (0), a random d -dimensional vector $X \in \mathbb{R}^d$ is set to have an α -unimodal distribution (where $\alpha > 0$) if, for every bounded, non-negative, Borel measurable function g on $\mathbb{R}^d$, the quantity $t^\alpha \mathbb{E}[g(tX)]$ is monotonically non-decreasing for $t \in (0, \infty)$. We denote a random vector following such a distribution as $X \sim P_\alpha$. A fundamental characteristic of this definition is its nesting property: if a distribution is α -unimodal and $\alpha < \beta$ then it is inherently β -unimodal as well ($X \sim P_\beta$).

To bridge this abstract multivariate definition with the intuitive understanding of one-dimensional unimodality, Olshen and Savage established the Decomposition Theorem [14]. In one-dimensional statistics, Khintchine’s classical theorem [15] demonstrated that a real random variable X has a unimodal distribution if and only if it can be represented as $X \sim UZ$, where U is a uniform random variable on (0,1) and Z is independent of U . The Decomposition Theorem extends this to $\mathbb{R}^d$, stating that a multivariate vector $X \sim P_\alpha$ if and only if it is distributed as:

$$X \sim U^{1/\alpha} Z$$

where U is uniformly distributed on (0,1) and Z is a random vector independent of U .

Therefore, a scalar variable follows an α -unimodal distribution if and only if X^α is unimodal according to the standard 1D definition.

This robust mathematical foundation gives rise to three pivotal properties of α -unimodality that are relevant to data transformation and clustering methodologies. The first is the translation property, according to which shifting a distribution across the feature space does not affect its unimodal nature. Formally, if we let $X \sim P_\alpha$ and consider a constant vector $c \in \mathbb{R}^d$, then the translated vector maintains the property, such that $X + c \sim P_\alpha$.

The second characteristic is the norm property. As formally stated [26], if a multidimensional random vector follows an α -unimodal distribution, its vector norm follows an α -unimodal distribution as well. Mathematically, if $X \sim P_\alpha$, then $\|X\| \sim P_\alpha$. This demonstrates that the distance magnitude of the data points from the mode preserves the unimodal structure, a concept fundamental for distance-based statistical tests.

Finally, the third characteristic is the projection property. This property provides a strict mathematical guarantee that linear projections of an α -unimodal distribution onto a lower-dimensional subspace will inherently preserve a form of unimodality. Specifically, for any given real matrix A from $\mathbb{R}^d$ to $\mathbb{R}^q$ and if $X \sim P_\alpha$ then its projection remains α -unimodal, expressed as: $AX \sim P_\alpha$.

Among these characteristics, the projection property is crucial for multidimensional clustering frameworks. It mathematically assures that if a high-dimensional dataset is genuinely α -unimodal, its structural integrity and single-peaked nature will theoretically be preserved even when the data is projected onto a single one-dimensional axis. This specific theorem forms the exact theoretical bedrock upon which modern multivariate tests, such as mud-pod, justify the translation of complex, high-dimensional data coordinates into one-dimensional manifold projections.

Based on these, as formally established in recent literature [26], if we consider a multidimensional vector $X \sim P_\alpha$ with a well-defined covariance matrix Σ and an arbitrary reference point $o \in \mathbb{R}^d$, then the distribution of the Mahalanobis distances with respect to o , given by $\sqrt{(x - o)^T \Sigma^{-1} (x - o)}$ is α -unimodal as well.

This specific lemma is of paramount importance. It guarantees that by measuring the Mahalanobis distance of all data points from a specific reference origin (or else named “observer”), the resulting one-dimensional distance perfectly retains the unimodal or multimodal nature of the original high-dimensional space. This provides the theoretical justification for utilizing Mahalanobis distances within multidimensional unimodality tests and allowing them to safely collapse complex feature spaces into one-dimensional distance distributions without losing critical structural information.

2.3.3 Random Projections and Dimensionality Reduction

A mathematically rigorous approach to handle multidimensional data without losing its structural integrity relies on random projections. The foundational theory enabling this is the Johnson-Lindenstrauss (JL) Lemma [16]. The JL lemma states that a set of points in high-dimensional space can be linearly projected into a space of much lower dimension in such a way that the distances between the points are preserved.

Formally, let $S := \{x_i\}_{i=1}^k$ be a subset of $\mathbb{R}^d$ and $\epsilon > 0$. The lemma establishes that if we construct a random projection matrix $\Pi \in \mathbb{R}^{d \times q}$, where $q \geq 8 \log(k)/\epsilon^2$ and its entries are independent and identically distributed as $\Pi_{ij} \sim \mathcal{N}(0, 1/d)$, distance preservation is mathematically guaranteed. Specifically, with a probability of at least $1/k$, for any pair of points $x_i, x_j \in S$, the following condition holds:

$$(1 - \epsilon)\|x_i - x_j\|^2 \leq \|\Pi x_i - \Pi x_j\|^2 \leq (1 + \epsilon)\|x_i - x_j\|^2$$

Furthermore, the practical application of the JL lemma requires a probabilistic guarantee for selecting an appropriate projection matrix. Let $\mathcal{R}_\Pi$ denote the family of random matrices that satisfy the distance preservation criteria specified by the lemma. As mentioned above, the probability that Π successfully preserves pairwise distances is bounded from below by $P(\Pi \in \mathcal{R}_\Pi) \geq 1/k$.

While the standard formulation of the JL lemma applies to the Euclidean distance, its extension to the Mahalanobis distance can be mathematically established by employing a square root decomposition of the covariance matrix and thereby generalize the distance-preserving properties of the JL lemma [17]. Beyond providing a strong theoretical foundation, this methodology addresses a limitation of high-dimensional statistical analysis, which is the singularity issue inherent in inverting the empirical covariance matrix [18], [19]. In high-dimensional spaces the covariance matrix becomes singular, preventing stable inversion. By projecting the data into a lower-dimensional subspace via Π we effectively mitigate this problem, rendering the estimation of Mahalanobis distances computationally stable and highly efficient.

2.4 Graph Theory Fundamentals

We will now establish the fundamental concepts of graph theory required for better understanding the algorithms. A weighted undirected graph $G = (V, E)$ is endowed with a set of vertices V and a set of edges E . Each edge connects an unordered pair of vertices and the associated edge weight represents the strength of this connection. The overall graph weight is defined as the sum of all weights. A subgraph $G' \subseteq G$ is a graph formed by a subset of vertices $V' \subseteq V$ and a subset of edges $E' \subseteq E$ connecting them. A path between two vertices $V_i, V_j \in V$ is a subgraph defined by a sequence of edges connecting starting from V_i and ending V_j . A path contains no cycles, as there is no repetition of vertices and therefore no repetition of edges in the sequence.

A graph G is considered connected if there exists a path between every pair of its vertices. Its maximal connected subgraphs are referred to as its connected components. A specific type of connected acyclic graph is a tree $T = (V, E)$ which satisfies the condition $|E| = |V| - 1$. Extending this concept, a forest F is an acyclic graph composed of a collection of trees, which could possibly be disconnected.

Any tree subgraph T that spans over all the vertices of G is called a spanning tree. A minimum spanning tree (MST) is a spanning tree with minimal graph weight, noting

that a single graph may have more than one MST. Respectively, one can define the spanning forest, or the minimum spanning forest (MSF), of a graph. Given $G = (V, E)$, an MSF can be efficiently computed in $\mathcal{O}(|E|\log|V|)$ time by executing the classical Kruskal’s algorithm [\[20\]](#). As it will be detailed in the following sections, the MSF is a mathematical object that is central to the way UniForCE, and our proposed methodology, discovers and represents data clusters.

Chapter 3. Foundational Methodologies

3.1 UniForCE

3.1.1 Locally Unimodal Cluster

As we mentioned in previous sections, estimating the optimal number of clusters k is considered quite a difficult problem. The main reason is because most of the traditional clustering algorithms often require k as an input or rely on specific cluster shape assumptions, such as Gaussianity. These assumptions frequently fail when dealing with real-world datasets. To address these structural limitations, the Unimodality Forest for Clustering Estimation (UniForCE) introduces a divergence by relying on the concept of local unimodality. Instead of enforcing unimodality across an entire cluster, which could restrict the cluster to a simple shape, UniForCE adopts the flexible “locally unimodal cluster” definition. A locally unimodal cluster extends across the data space for as long as the unimodality property is locally preserved across pairs of adjacent subregions. This definition allows for the identification of highly irregular geometries, such as arcs, nested rings and complex non-linear shapes, without requiring ad-hoc hyperparameters or data transformations into alternative vector spaces.

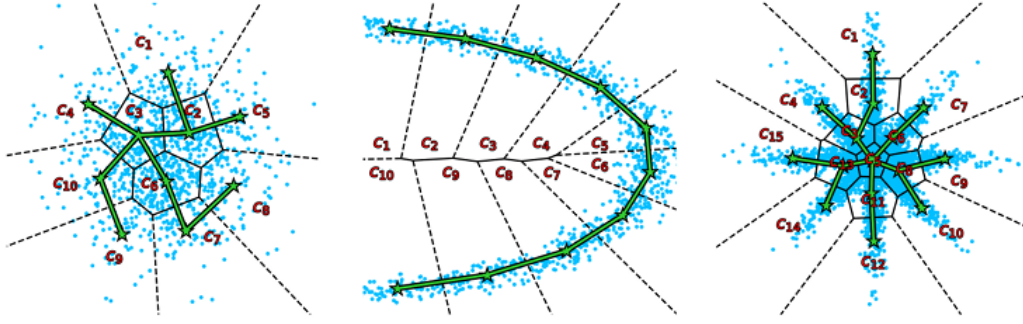


Figure 3.1 Examples of locally unimodal clusters.

Fig. 3.1 [28] illustrates examples of locally unimodal clusters, where each c_i is a subcluster and an edge between subclusters indicates that their union is unimodal.

The UniForCE methodology operates in a bottom-up, agglomerative fashion and comprises three distinct computational phases: Overclustering, Unimodal Pair Testing and Clustering by Subcluster Aggregation.

Algorithm 1 The UniForCE general framework for locally unimodal clustering

Require: X (dataset), K' (number of subclusters, $K' \gg k^*$), M (minimum number of subclusters), α (significance level)

- 1: **Overclustering:** Compute an initial overclustering of X into K' homogeneous subclusters lying in convex subregions. Eliminate small subclusters with less than M datapoints and determine the final overclustering partition C^+ into $K \leq K'$ subclusters.
 - 2: **Unimodal Pair Testing:** Apply a statistical test to determine whether the union of two subclusters admits unimodality (with significance level α).
 - 3: **Clustering by subcluster aggregation:** Compute the final clustering partition C by determining the connected components of the unimodality graph G of the overclustering C^+ .
 - 4: **return** the locally unimodal clustering partition C .
-

3.1.2 Overclustering

The initial phase of the algorithm aims to segment the raw data space into a set of small, highly homogeneous subregions. Given a dataset X containing N data points in a d -dimensional space $\mathcal{R}^d$, UniForCE computes an initial overclustering partition $\mathcal{C}^+$ consisting of K' subclusters, where K' is a user defined integer significantly larger than the true (and unknown) number of clusters k^* . It is a strict geometric requirement that these subclusters reside in convex subregions to accurately represent localized data densities.

To achieve this, the standard k-means algorithm is often insufficient for large values of K' due to its sensitivity to random initialization. For this reason, UniForCE employs the global k-means++ algorithm [21], an incremental variant that produces highly robust partitions by solving intermediate clustering subproblems with the addition of one cluster each time and the selection of candidate center positions using the k-means++ seeding procedure [22].

Following the initial partitioning, a preprocessing step is executed to enforce statistical reliability in the subsequent statistical tests. Subclusters containing fewer data points than a specified minimum threshold M (typically set to $M = 25$) are eliminated. The data points belonging to these eliminated clusters are immediately redistributed to the remaining valid subclusters based on the standard k-means minimum distance assignment rule. This procedure ensures that all remaining subclusters in the final overclustering partition possess sufficient sample sizes for valid density analysis.

3.1.3 Unimodal Pair Testing

The fundamental step of UniForCE is the unimodal pair testing procedure. This module assesses whether the geometric union of two adjacent subclusters, denoted as c_i and c_j , forms a cohesive, unimodal structure. The underlying hypothesis is that if a significant density gap exists between the two subclusters, their union is rendered multimodal, indicating that they belong to distinct, separate clusters.

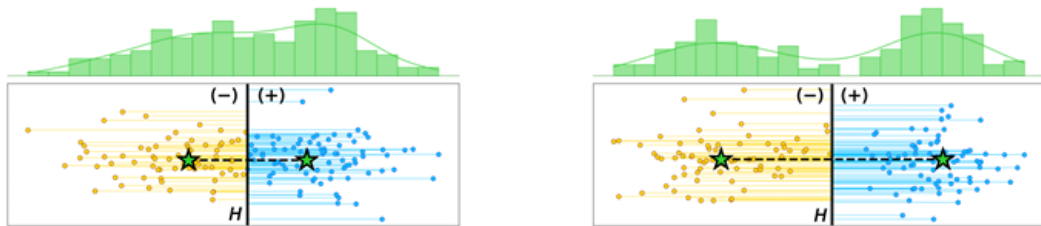


Figure 3.2 [28] Unimodal Pair testing. Two subclusters c_i and c_j , appear in blue and yellow, respectively, and their centers are shown as stars. The dotted line connects the two centers, while the rigged line is its perpendicular bisecting hyperplane H_{ij} . Histograms present the density of the univariate set P_{ij} , with the point-to-hyperplane signed distances, which is tested for unimodality using the dip-test. (a) Unimodal case: No density gap is observed between the subclusters and P_{ij} is decided as unimodal. (b) Multimodal case: A considerable density gap is observed between the subclusters and P_{ij} is decided as multimodal.

To perform this evaluation, UniForCE relies on a geometrically defined one-dimensional projection. Let μ_i and μ_j represent the geometric centroids of subclusters c_i and c_j , respectively. The algorithm defines the connecting vector $r_{ij} = \mu_i - \mu_j$ and establishes the perpendicular bisecting hyperplane H_{ij} passing precisely through the midpoint of r_{ij} . Then, the algorithm computes the univariate set of signed distances, P_{ij} , for all data points within the union $c_i \cup c_j$ with respect to the hyperplane H_{ij} .

We then apply Hartigan’s dip-test [13] to the extracted set P_{ij} , against a defined statistical significance level α . However, the standard dip-test can be vulnerable to false negatives, failing to reject unimodality, when the local modes are highly imbalanced. This typically occurs when one subcluster contains a considerable amount of more points than the other. To mitigate this statistical bias, UniForCE incorporates a balanced Monte Carlo subsampling strategy. The algorithm determines the minimum cardinality between the two subclusters, $s = \min(|c_i|, |c_j|)$. It then draws a uniform random subsample of exactly s from the larger subcluster and merges it with the entirety of the smaller subcluster. This perfectly balanced set is evaluated using the dip-test. To ensure robustness, this random subsampling and testing process is repeated for an odd number of Monte Carlo simulations (e.g., $L=11$). The final outcome of whether the pair is unimodal or multimodal is determined by a simple majority vote among the L simulation results.

3.1.4 Subcluster Aggregation and Spanning Forest

Once the local unimodality between pairs of subclusters can be evaluated, the algorithm proceeds to infer the global macro-structure of the dataset. This is established through the abstract construction of a “unimodality graph” G , where the vertices correspond to the individual subclusters and the undirected edges represent unimodal pairs. A distinct path within this graph signifies a sequence of subclusters where any two successive nodes form a unimodal pair. Therefore, the connected components of the unimodality graph directly and optimally correspond to the maximal locally unimodal clusters.

To optimize computational efficiency and avoid performing the exhaustive $\frac{K(K-1)}{2}$ unimodality tests required for constructing a complete graph, UniForCE approximates the connected components by computing a Unimodality Spanning Forest. The algorithm initializes a complete distance graph where edge weights correspond to the Euclidean distances between the respective subcluster centroids. Operating under the intuitive assumption that unimodality typically extends between geometrically proximate subregions, the algorithm evaluates pairs of subclusters in ascending order of their spatial distance. This is computationally realized via a modified Kruskal's algorithm [23].

The spanning forest is initialized with K disconnected, singleton trees, each representing one initial subcluster. As the distance-sorted edges are sequentially traversed, the algorithm first checks if the corresponding pair of vertices already belongs to the same tree in the forest. If they belong to different trees, the rigorous unimodal pair test is executed. If the test successfully confirms unimodality, an edge is added to the forest, essentially merging the two sub-trees. This greedy approximation minimizes the number of required, computationally heavy statistical tests without compromising or altering the final topological clustering partition. The algorithm terminates naturally when all relevant edges have been traversed.

3.1.5 Computational Complexity

Despite its statistical rigor, the overall computational complexity of the UniForCE methodology remains highly competitive for robust density-based clustering. The initial overclustering phase utilizing global k-means++ [21] requires $\mathcal{O}(QKNd)$ time, where Q represents a small constant number of candidate centers examined per iteration. The subsequent preprocessing and structural redistribution take $\mathcal{O}(KN)$ time. The dominant computational cost arises from evaluating the dip-statistic via the unimodal pair tests across the distance-ordered graph, requiring $\mathcal{O}(N \log N)$ time. Finally, the topological construction of the spanning forest operates in $\mathcal{O}(K^2 \log K)$ time. Summing these individual components, the total theoretical time complexity of the UniForCE algorithm is bounded by $\mathcal{O}(KNd + KN + N \log N + K^2 \log K)$.

3.2 Mud-pod

3.2.1 Randomization Hypothesis

When we want to assess the unimodality of datasets composed of multidimensional data vectors, inherently a lot of challenges arise. The mud-pod method overcomes these limitations by transforming the problem into a progressively simpler and more Gaussian-like space using linear random projections.

The fundamental concept relies on selecting an arbitrary data point, termed the observer, and calculating its distances to all other points within the subset. This procedure captures a snapshot of the underlying cluster morphology. In the presence of a single, highly homogeneous cluster, the distribution of these point-to-point distances is expected to be unimodal. The narrative and structural perspective presented by this distance distribution is highly contingent upon the special location of the chosen observer. By pairing a specific random projection with an observer's perspective, it formulates what is formally termed as view.

Mathematically, given a set S_X of points drawn from a distribution $X \sim P_\alpha$, let $o \in S_X$ represent a randomly selected observer. We define the set of Mahalanobis distances with respect to this observer o as:

$$\mathcal{D}_S^o = \left\{ \left\| \Sigma^{-\frac{1}{2}}(x - o) \right\| \mid x \in S_X \setminus \{o\} \right\}$$

Subsequently, applying a distance-preserving random projection matrix $\Pi \in \mathcal{R}_\Pi$, we define the set of Mahalanobis distances for the randomly projected points with respect to the observer o as:

$$\Pi \circ \mathcal{D}_S^o = \left\{ \left\| \Sigma_\Pi^{-\frac{1}{2}} \Pi(x - o) \right\| \mid x \in S_X \setminus \{o\} \wedge \Pi \in \mathcal{R}_\Pi \right\}$$

Where $\Sigma_\Pi = \Pi \Sigma \Pi^T$. Because the original distribution is assumed to be α -unimodal ($X \sim P_\alpha$), the elements within both $\mathcal{D}_S^o$ and its projected counterpart $\Pi \circ \mathcal{D}_S^o$ retain α -unimodal distributions.

As such, we have the following observations, called Randomization Hypothesis (RH). According to this, given $X \sim P_\alpha$, the distributions of elements within $\mathcal{D}_S^o$ and $\Pi \circ \mathcal{D}_S^o$ retain α -unimodality under any transformation $\Pi \in \mathcal{R}_\Pi$. Essentially, the RH states that, under the assumption of multivariate unimodality, every set of randomly projected distances should exhibit unimodality. Consequently, any detected deviation from this expected

unimodal behavior across different views serves as a strong signal of multimodality within the original data distribution.

The incorporation of random projections confers several advantages for the clustering verification. Firstly, random projections maintain pairwise distances, as endorsed by the JL lemma. Secondly, they transform the complex high-dimensional challenge into a lower-dimensional space that heavily resembles a mixture of Gaussians, conforming to the Diaconis-Freedman effect. Another interesting advantage is that the projections ameliorate singularity issues that frequently arise when inverting the covariance matrix required when using the Mahalanobis distances. Lastly, they establish the mathematical foundation necessary for harnessing Monte Carlo simulations for hypothesis testing [24], allowing us to sample from a distribution ($\mathcal{R}_\Pi$) that is computationally simpler than the original dataset.

3.2.2 The mud-pod Algorithm

Building upon the aforementioned principles, the mud-pod test formalizes the verification of an individual cluster. For a specified unimodality index α and a subset of points S_X representing the cluster, we define the core statistical hypothesis:

$$H_0: X \sim P_\alpha \quad \text{vs} \quad H_1: X \not\sim P_\alpha$$

We also define the pairing of a random projection Π with an observer o as a random view. Under the null hypothesis, the derived set of distances $\Pi \circ \mathcal{D}_S^o = \{d_i\}_{i=1}^{N-1}$ is α -unimodal. By exponentiating these distances, the set $\{d_i^\alpha\}_{i=1}^{N-1}$ becomes strictly unimodal in the classical sense, which allows the direct application of the one-dimensional Hartigan's dip-test.

Let $T(\Pi \circ \mathcal{D}_S^o)$ denote the p-value derived from the dip-test for a single random view. Given a predefined statistical significance level $\alpha \in [0,1]$, the null hypothesis is rejected for that specific view if and only if $T(\Pi \circ \mathcal{D}_S^o) \leq \alpha$. Because a single view might not provide a definite conclusion, the algorithm relies on a Monte Carlo simulation framework, leveraging the Monte Carlo hypothesis testing theory [24], utilizing a series of M random views, comprised of M random projections $\{\Pi_i\}_{i=1}^M \subset \mathcal{R}_\Pi$. The rationale dictates that as more random views reject the null hypothesis, the confidence regarding the underlying multimodality of the data subset increases.

Based on the fact that the dip-statistic has a well-defined c.d.f. [13], the c.d.f. of the dip-test, $J_N(t) = P(T(\Pi \circ \mathcal{D}_S^o) \leq t | S_X)$, is explored. The aforementioned probability can be approximated empirically over the joint distribution of the M random views. Using the indicator function $I\{ \cdot \}$, the empirical c.d.f. $\hat{J}_{N,M}(t)$ is defined as:

$$\hat{J}_{N,M}(t) = M^{-1} \sum_{i=1}^M I\{T(\Pi \circ \mathcal{D}_S^o) \leq t\}$$

According to the Glivenko-Cantelli theorem, this approximation converges with probability 1 to the true c.d.f. $J_N(t)$ as the number of simulations increases [25]. Notably, the Dvoretzky, Kiefer, Wolfowitz inequality provides bounds on the closeness between $\hat{J}_{N,M}(t)$ and $J_N(t)$ for a given M . Specifically, we have:

$$P\left(\sup_{t \in \mathbb{R}} |\hat{J}_{N,M}(t) - J_N(t)| > \tau\right) \leq C e^{-2M\tau^2}$$

While observers could theoretically be sampled uniformly at random, empirical evaluations suggest a highly optimized strategy: selecting observers from a specific upper percentile of the distance distribution from the sample mean. Specifically, points situated farther away from the central mean exhibit a superior topographical capability to discern the “elevations” or density gaps formed by the presence of distinct clusters [12]. Thus, in the algorithm, the observer o is uniformly sampled from the p -th percentile of the data.

The pipeline of the mud-pod is described by the following sequence of operations, iteratively executed M times. First, the multi-dimensional data points are linearly projected into a lower-dimensional subspace via a randomly generated projection matrix Π_i , resulting in the projected data $X\Pi_i$. Then an observer point o is selected from the p -th percentile of the projected Mahalanobis distances relative to the projected subset’s mean and the set of one-dimensional Mahalanobis distances from the observer o to all other projected points, denoted as $\Pi_i \circ \mathcal{D}_S^o$, is computed. Finally, Hartigan’s dip-test is conducted on the exponentiated distances to assess one-dimensional unimodality, yielding a binary rejection indicator based on the significance threshold α .

After concluding the M Monte Carlo simulations, the final multivariate p -value is calculated as the proportion of views that resulted in a rejection of the null hypothesis:

$$p_{value} = \frac{1}{M} \sum_{i=1}^M I\{T(\Pi \circ \mathcal{D}_S^o) \leq \alpha\}$$

If this aggregated p_{value} falls below the predetermined threshold, the set of points examined, i.e. a cluster, is deemed multimodal. Algorithm 2 details the mud-pod test pipeline.

Algorithm 2 mud-pod($\alpha, X, \alpha, M, p, \epsilon$)

Input: α (the positive unimodality index), X (a set of real vectors), α (a significance threshold), M (number of simulations), p (p -th percentile), ϵ (distance distortion)

Output: p -value of the test

- 1: **for** $i = 1$ to M **do**
 - 2: Project the points via a $\left\lceil \frac{8 \log(|X|)}{\epsilon^2} \right\rceil$ random projection, resulting in $X\Pi_i$.
 - 3: Select an observer $\mathbf{o}$ from the p -th percentile of the projected Mahalanobis distances from the mean.
 - 4: Compute the set of distances from $\mathbf{o}$, i.e. $\Pi \circ \mathcal{D}_S^{\mathbf{o}}$.
 - 5: Conduct a dip test on exponentiated $\Pi \circ \mathcal{D}_S^{\mathbf{o}}$ distances.
 - 6: **end for**
 - 7: **return** $p_{value} = \frac{1}{M} \sum_{i=1}^M I\{T(\Pi \circ \mathcal{D}_S^{\mathbf{o}}) \leq \alpha\}$
-

Chapter 4. The Proposed Methodology

4.1 Rationale of the Proposed Architecture

The UniForCE offers a flexible framework for identifying locally unimodal clusters through the bottom-up aggregation of subclusters. However, its foundational step relies on partitioning the dataset into homogeneous and convex subregions using the global k-means++ algorithm. While it is effective for geometric partitioning, this approach lacks a statistical guarantee that the resulting subclusters are unimodal and subsequently homogeneous. To address this specific gap, our proposed methodology enhances the UniForCE pipeline by integrating the mud-pod test. By having mud-pod as a verification mechanism directly after the overclustering phase, our pipeline ensures that every subcluster is mathematically validated as unimodal before any aggregation occurs, thereby reinforcing the statistical integrity of the entire clustering process.

4.2 The Enhanced Pipeline

In our proposed methodology, the dataset is initially over-segmented into a predefined number of subclusters utilizing the global k-means++ algorithm. Immediately after this partitioning, the proposed contribution introduces the multivariate verification step. During this stage, the mud-pod test is independently applied to each generated subcluster. The algorithm evaluates the null hypothesis that the data distribution within this specific subcluster is unimodal.

If the null hypothesis is accepted, the subcluster is validated as unimodal and retained for the subsequent structural phase. On the other hand, if the test rejects the null hypothesis, indicating the presence of multimodality, the problematic subcluster must

undergo a corrective action to enforce homogeneity. This is achieved by applying the K-means algorithm on the specific subcluster, in order to split it into two new and more homogeneous subclusters. Of course, after this we have to apply the mud-pod test on the newly generated subclusters to test their unimodality. It should be noted that if the number of points inside the subcluster is less than M (the minimum threshold of the algorithm) then we do not apply the statistical test on this cluster.

Once all subclusters are statistically verified and corrected, the algorithm eliminates the clusters with fewer than the threshold M points and redistributes these data points to the remaining valid subclusters. After that we transition to the pair testing phase. We assess whether the union of adjacent subclusters preserves local unimodality by applying the Hartigan's dip-test. At the end, a minimum spanning forest is constructed over the resulting unimodality graph, and the connected components of this graph are extracted to form the final, maximal locally unimodal clusters.

Algorithm 3 The proposed methodology (SMCUT)

Require: X (dataset), K' (number of subclusters, $K' \gg k^*$), M (minimum number of subclusters), α (significance level)

- 1: **Overclustering:** Compute an initial overclustering of X into K' homogeneous subclusters lying in convex subregions.
 - 2: **Unimodality Validation:** Apply the mudpod test to every subcluster and split further every multimodal subcluster. Eliminate small subclusters with less than M datapoints and determine the final overclustering partition C^+ into $K \leq K'$ subclusters.
 - 3: **Unimodal Pair Testing:** Apply a statistical test to determine whether the union of two subclusters admits unimodality (with significance level α).
 - 4: **Clustering by subcluster aggregation:** Compute the final clustering partition C by determining the connected components of the unimodality graph G of the overclustering C^+ .
 - 5: **return** the locally unimodal clustering partition C .
-

Chapter 5. Experiments and Evaluation

5.1 Introduction to the Experiments

In this chapter, we present a comprehensive experimental evaluation of our proposed algorithm. The primary objective is to benchmark its clustering performance against the baseline UniForCE framework. The chapter is structured as follows: First, the experimental setup is detailed, outlining the diverse collection of real-world and synthetic datasets, the evaluation metrics and the implementation details. Subsequently, a thorough comparative analysis is conducted on both real-world and synthetic datasets. Furthermore, we investigate the algorithm's optimal overclustering resolution (hyperparameter K) and evaluate the comparative performance of both methods at their respective optimal configurations. Finally, the chapter concludes with a critical analysis of the algorithm's limitations, identifying specific failure cases and the theoretical constraints that cause them.

5.2 Experimental Setup

5.2.1 Datasets

[Table 1](#) summarizes the datasets that were used for the experimental evaluation, which vary in size N , dimensions d , number of clusters k , data type and domain of origin. All datasets used were subjected to a preprocessing step involving Min-Max normalization. This technique maps the attributes of data points to the $[0,1]$ interval, preventing in that way attributes with large values dominating the distance calculations.

Dataset	Type	Description	N	d	k
Waveform-v1	Vector	Waveforms with multiple attributes	5000	21	3
Mice-Protein Expression	Tabular	Expression level of proteins	1080	77	8
Optdigits	Image	Handwritten digits	5620	8×8	10
Pendigits	Time-series	Handwritten digits	10992	16	10
EMNIST Balanced Digits	Vector	Handwritten digits	28000	10	10
EMNIST MNIST	Vector	Handwritten digits	70000	10	10
EMNIST Balanced Letters	Vector	Handwritten letters (A-J)	28000	10	10
Isolet	Spectral	Speech recordings pronouncing letters	7797	617	26
TCGA	Tabular	Cancer gene expression profiles	801	20531	5

Table 1: The real datasets used in our experiments, N is the number of data instances, d is the dimensionality and k denotes labeled classes that we consider as the ground truth number.

5.2.1.1 Real-World Datasets

The Optdigits dataset is a classical machine learning benchmark for optical character recognition, consisting of normalized bitmaps of handwritten digits. The dataset comprises $N = 5620$ data points. Each instance represents an 8×8 pixel image, resulting in a relatively low-dimensional feature space of $D = 64$. The ground-truth labels are divided into $K=10$ distinct classes, corresponding to the digits ‘0’ through ‘9’.

The Pendigits dataset captures the dynamic trajectory of pen strokes during the writing of digits. It consists of $N=10992$ instances. The data is represented as a sequence of (x,y) coordinates, in a 16-dimensional feature space. Similar to other datasets with digits, it contains $K=10$ classes.

The EMNIST (Extended MNIST) datasets provide a more challenging and modern alternative to traditional digit recognition datasets. For this evaluation, the EMNIST-MNIST and EMNIST Balanced Digits (EMNIST-BD) were utilized.

The raw pixel-space complexity of these images is exceptionally high. Consequently, we utilized the pre-processed, high-quality data embeddings provided by the authors of the original UniForCE paper [28]. As detailed in their methodology, these embeddings were extracted using an Autoencoder (AE) architecture consisting of a 3-layer convolutional encoder and a two-layer MLP and the decoder part is symmetrical to the encoder. It was trained for 100 epochs using the Adam optimizer. Therefore, for both EMNIST MNIST

($N=70000$ and $K=10$) and EMNIST-BD ($N=28000$ and $K=10$), our clustering evaluation is performed directly on these latent representations, while successfully compressing the feature space to $D = 10$ dimensions.

Shifting from numeric to alphabetical morphological structure, the EMNIST-BL dataset focuses on handwritten letters, providing a significantly different challenge. This specific subset contains a total of $N = 28000$ data points, filtered to include $K = 10$ classes representing character from the letter 'A' to 'J'. Consistent with the methodology described previously, we operated on the pre-extracted $D=10$ -dimensional latent space embeddings provided by the original UniForCE study and generated via their AE architecture.

The ISOLET dataset represents a shift to acoustic signal processing, containing audio recordings of spoken letters. It comprises of $N = 7797$ instances. The acoustic signals are transformed into continuous feature vectors spanning $D = 617$ dimensions, categorized into $K = 26$ classes (one for each of the letter in the English alphabet).

The Cancer Genome Atlas (TCGA) dataset provides a complex bioinformatics scenario, consisting of gene expression profiles obtained via RNA sequencing from various cancer samples. It is a high-dimensional biological dataset containing 801 data instances. It maps to $K = 5$ distinct classes, representing different cancer type pathways. The feature space includes clinical information and normalized gene annotations, leading to a vast dimensionality of $D = 20531$.

Waveform-v1 is a synthetically generated benchmark designed to simulate complex wave interactions and noise interference. It encompasses $N = 5000$ data points. The instances are distributed among $K = 3$ generated wave classes, evaluated across a relatively low-dimensional space of $D = 22$.

The Mice-Protein Expression dataset captures real-world biological profiles, measuring detectable signals in the nuclear fraction of the cortex. It includes $N = 1080$ data points. The features represent the expression levels of $D = 77$ proteins or protein modifications. The samples are divided into $K = 8$ classes based on a combination of genotype, behavior and treatment features.

5.2.1.2 Synthetic Datasets

In addition to real-world datasets, a carefully selected series of synthetic two-dimensional benchmark datasets was utilized. While the real-world datasets evaluate the algorithm's performance on highly dimensional and noisy feature spaces, synthetic datasets are crucial for isolating and testing specific topological vulnerabilities. These 2-

dimensional datasets are generated to form distinct and challenging geometric shapes, allowing for a precise visual evaluation of the clustering process.

To examine the algorithm’s performance on non-convex regions, the banana dataset was used, consisting of two moons. It has a total of $N = 4811$ data points, with two ground-truth labels and it also has a narrow margin of separation. The next dataset is the Complex9, which consists of $N = 3031$ unique points and $K=9$ unique labels.

Another interesting dataset utilized is the smile3, which illustrates a smiling face, with $N=1000$ data points and $K=4$ classes. Furthermore, the twodiamonds dataset was used, consisting of $N = 800$ data points and $K = 2$ classes while another 2-dimensional dataset named Complex2D was used with $N=5500$ points and $K=6$ unique labels.

Lastly, a 3-dimensional dataset called chainlink was used, containing $N=1000$ data points and $K=2$ classes.

5.2.2 Evaluation Metrics

For evaluating the clustering partition and how it matches the ground-truth label information, we compute the Adjusted Mutual Information (AMI) measure that takes values from 0 to 1 and is defined as:

$$AMI(U, V) = \frac{MI(U, V) - E[MI(U, V)]}{AVG(H(U), H(V)) - E[MI(U, V)]}$$

where U is the vector of ground-truth labels, V denotes the vector of cluster labels produced by a clustering algorithm, MI is the Mutual Information measure, $H(\cdot)$ is the entropy of each partition and $E[\cdot]$ is the expected value. AVG is the method used to compute the normalizer in the denominator which is set to ‘arithmetic’. Besides this metric we also consider the estimated number of clusters k .

5.2.3 Implementation Details and Baselines

To evaluate the effectiveness of the proposed methodology, the original UniForCE algorithm is utilized as the primary baseline. The two algorithms were evaluated using static hyperparameter configurations inspired by the UniForCE implementation. Specifically, for the UniForCE components, the minimum subcluster size was set to $M = 25$ data points, the significance level to $\alpha = 0.001$ and the number of Monte Carlo simulations for the paired dip-test to $L = 11$.

For the mud-pod verification step in our proposed variant, the static hyperparameters were set as follows: $\alpha = 0.05$, the observer type as percentile, the distance type as

Mahalanobis, the number of simulations to 20 and the projection type to Johnson and Lindenstrauss.

To perform these experiments, we used the Jupyter Notebook environment. To ensure a fair comparison between the baseline UniForCE and our proposed version, SMCUT, we maintained identical initial conditions for both methods. Specifically, the initial overclustering step is executed once for both algorithms, so they can have the same overclustering. The resulting subclusters were then passed through our function that performs the unimodality tests on these clusters.

Subsequently, the final Kruskal-based aggregation phase was executed twice independently: once using the raw overclustering output, thus simulating the UniForCE pipeline, and once using the output from the unimodality validation function to represent our enhanced approach. This shared initialization isolates the impact of the mud-pod component, guaranteeing that any divergence in the final clustering performance is strictly attributable to the validation step rather than stochastic variations during the initial overclustering.

5.3 Analysis on Real-World Datasets

	<i>Optdigits (10)</i>				<i>Pendigits (10)</i>				<i>EMNIST-BD (10)</i>			
	Init	Uni	k	AMI	Init	Uni	k	AMI	Init	Uni	k	AMI
<i>Uniforce</i>	100	-	8	0.8701	100	-	17	0.8513	100	-	12	0.8778
	80	-	8	0.8288	80	-	17	0.8401	80	-	10	0.8859
	50	-	14	0.8627	50	-	18	0.8332	50	-	10	0.8727
	30	-	13	0.8235	30	-	15	0.7954	30	-	9	0.8349
	15	-	12	0.7960	15	-	13	0.7475	15	-	12	0.7556
<i>SMCUT</i>	100	109	5	0.6551	100	110	16	0.8283	100	101	11	0.8764
	80	81	8	0.8273	80	84	17	0.8378	80	80	10	0.8859
	50	53	13	0.8410	50	54	18	0.8211	50	50	10	0.8727
	30	30	13	0.8235	30	34	16	0.7971	30	30	9	0.8349
	15	16	13	0.8022	15	19	14	0.7611	15	15	12	0.7556

Table 2: Experiments on the datasets Optdigits, Pendigits, EMNIST-BD datasets

	<i>EMNIST-BL (10)</i>				<i>EMNIST-M (10)</i>				<i>Waveform-v1 (3)</i>			
	Init	Uni	k	AMI	Init	Uni	k	AMI	Init	Uni	k	AMI
<i>Uniforce</i>	100	-	9	0.7114	100	-	15	0.8749	100	-	3	1.0000
	80	-	9	0.6841	80	-	16	0.8572	80	-	3	1.0000
	50	-	10	0.7516	50	-	12	0.8686	50	-	3	1.0000
	30	-	13	0.7168	30	-	13	0.8349	30	-	3	1.0000
	15	-	10	0.6494	15	-	12	0.7763	5	-	3	0.7661
<i>SMCUT</i>	100	101	9	0.7114	100	100	15	0.8749	100	103	3	1.0000
	80	81	9	0.7123	80	80	16	0.8572	80	87	3	1.0000
	50	50	10	0.7516	50	50	12	0.8686	50	53	3	1.0000
	30	30	13	0.7168	30	30	13	0.8349	30	30	3	1.0000
	15	16	12	0.6483	15	15	12	0.7763	5	6	3	1.0000

Table 3: Experiments on the datasets EMNIST-BL, EMNIST-M and Waveform-v1

	<i>ISOLET (26)</i>				<i>Mice-Protein (8)</i>				<i>TCGA (5)</i>			
	Init	Uni	k	AMI	Init	Uni	k	AMI	Init	Uni	k	AMI
<i>Uniforce</i>	150	-	26	0.7209	50	-	9	0.9519	50	-	6	0.9242
	100	-	30	0.7237	40	-	9	0.9744	40	-	6	0.9217
	80	-	27	0.7228	30	-	9	0.9687	30	-	5	0.9715
	50	-	27	0.7258	20	-	9	0.9734	20	-	5	0.9715
	40	-	23	0.7078	14	-	8	1.0000	10	-	5	0.9496
<i>SMCUT</i>	150	161	25	0.7153	50	55	6	0.8375	50	51	6	0.9242
	100	105	27	0.7259	40	50	8	0.9695	40	40	6	0.9217
	80	82	29	0.7385	30	49	6	0.8838	30	30	5	0.9715
	50	51	25	0.7253	20	29	8	0.9887	20	20	5	0.9715
	40	42	23	0.7089	14	18	8	0.9902	10	11	5	0.9496

Table 4: Experiments on the datasets ISOLET, Mice-Protein and TCGA.

The three tables above summarize the performance of both UniForCE and SMCUT across the real-world datasets described in Section 5.2.1 for a variety of initial values of hyperparameter K.

The experimental results demonstrate that SMCUT consistently achieves higher or comparable AMI scores across most datasets and most K values. For instance, in the EMNIST-BL dataset for $K=80$ the proposed algorithm improved the score from 0.6841 to 0.7123 by identifying a multimodal initial subcluster and dividing it. Another notable example is the ISOLET dataset for $K=80$ which improves the AMI score from 0.7228 to 0.7385 by identifying 2 multimodal clusters at the overclustering step and dividing them. These improvements suggest that the integration of the mud-pod test effectively refines the initial overclustering resolution.

Regarding the estimation of the number of clusters (k), SMCUT shows a notable tendency to provide estimations closer to the ground truth labels. While the original UniForCE tends to overestimate or underestimate k when the initial subclusters are not perfectly homogeneous, our approach leverages the multivariate unimodality test which evaluates the Mahalanobis distances over random projections to ensure that each subcluster is a valid block for the final partition, enhancing the stability of the method.

The results indicate that SMCUT is substantially less sensitive to variations in K . This reduced dependency on the initial overclustering resolution is a major advantage of our approach. Because the mud-pod test rigorously verifies the statistical unimodality of the subclusters before any graph-based merging occurs, it acts as a safety mechanism. Consequently, even if K is set to a sub-optimal value, the algorithm successfully prevents incorrect cluster aggregations, remaining remarkably stable compared to the original UniForCE. An additional advantage of SMCUT is that it exhibits a deeper understanding of the local topology. Although some datasets result in similar overall performance (AMI and k) for both algorithms, our approach successfully identifies and prevents the aggregation of multimodal subclusters after the overclustering step, thanks to the early intervention of the mud-pod test. This early detection ensures a more robust and statistically sound foundation for the subsequent graph construction.

While SMCUT demonstrated superior or comparable performance across most of the evaluated datasets, there were specific instances where the proposed method yielded suboptimal results (e.g., in the Mice-Protein dataset). A comprehensive analysis of these failure cases and the underlying limitations of the algorithm is presented in Section 5.7.

5.4 Analysis on Synthetic Datasets

To gain a deeper understanding of the clustering behavior of SMCUT, a series of experiments were conducted using a variety of 2D and 3D synthetic datasets. These datasets are specifically chosen for their complex topological features, including non-convex shapes, nested structures and non-linearly separable clusters, which pose significant challenges for traditional clustering methods.

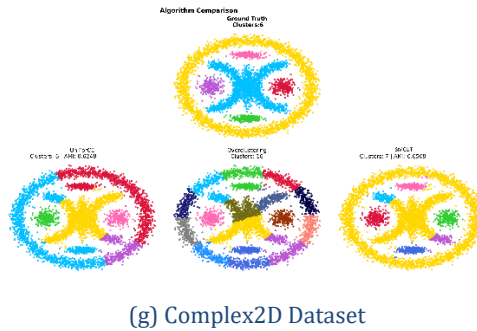
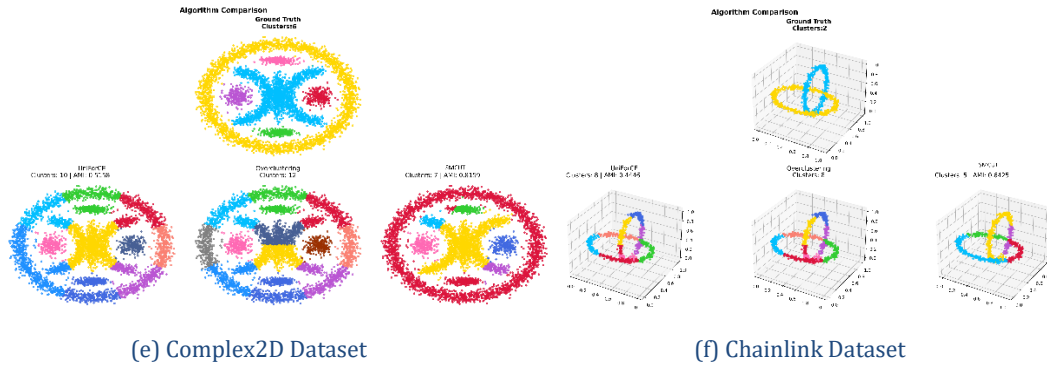
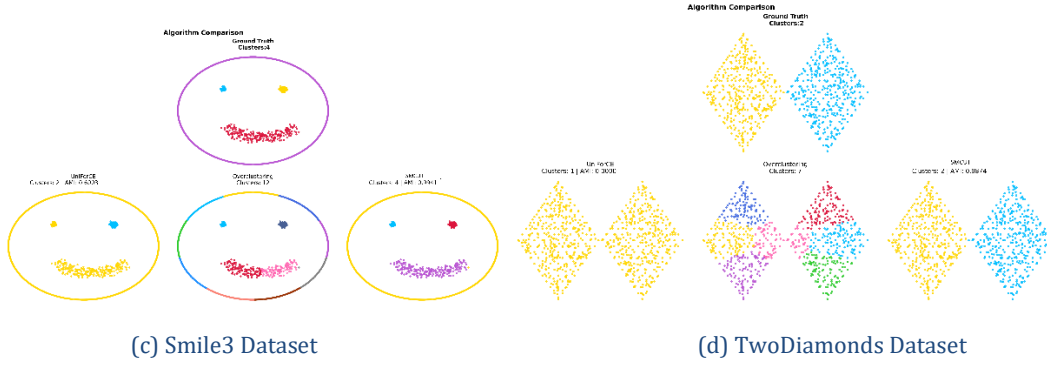
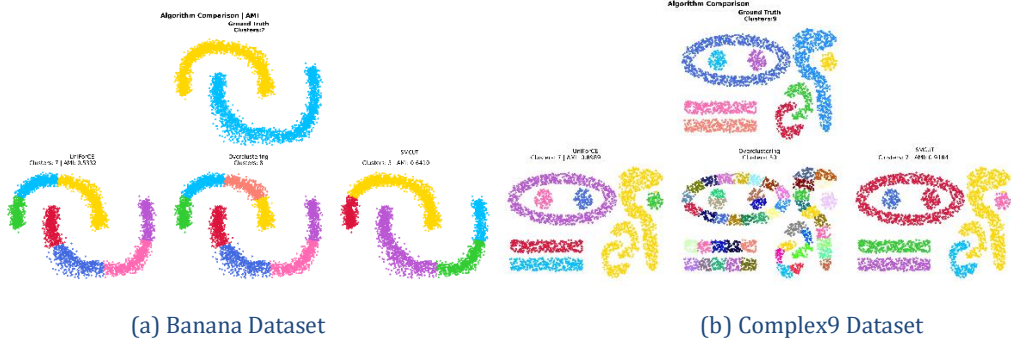


Figure 5.1: Clustering results on synthetic datasets. The top of each image shows the ground truth. The bottom row displays the UniForCE result on the left, the overclustering result in the middle and the SMCUT result on the right.

[Figure 5.1](#) illustrates the clustering solutions generated by the original UniForCE and the proposed SMCUT. Both algorithms successfully identify the underlying structures in many cases, demonstrating the versatility of the locally unimodal cluster definition. However, a closer inspection of the subcluster aggregation process reveals distinct qualitative differences.

Starting with the banana dataset in [Figure 5.1\(a\)](#) initialized with $K=8$ subclusters, the standard overclustering phase struggles to partition non-linear data curves into sufficiently small and convex regions. This limitation results in curved, non-homogeneous subclusters that fundamentally violate the core assumptions of the baseline UniForCE algorithm. Consequently, during the aggregation step the dip-test detects internal density gaps – indicating multimodality – and prematurely halts the merging process.

In contrast, SMCUT successfully mitigates this issue. By leveraging random projections, the mud-pod test detects this inherent multimodality at an early stage. This early recognition forces the problematic subclusters to split further (expanding from 8 to 11 subclusters), ultimately transforming them into smaller, more linear and strictly unimodal regions. This enables the construction of a highly accurate unimodality graph, yielding a significantly higher AMI and a much better estimation of the final cluster count.

In essence, the integration of the mud-pod test effectively relaxes the strict dependency on the initial convex shape assumption, rendering the overall UniForCE framework substantially more flexible and robust against poor initial overclustering.

Subsequent analysis on the Complex9 dataset, depicted in [Figure 5.1\(b\)](#), highlights a different aspect of the algorithm’s behavior. Observing the region with the two small, curved structures on the right side of the dataset, it is highly likely that the initial overclustering generated subclusters that obscured the underlying density gap. The application of the mud-pod algorithm successfully detected multimodality within these initial subclusters, triggering further splits. Consequently, the density gap became significantly more distinct during the unimodal pair testing phase, effectively preventing erroneous merging of these two distinct structures.

When examining the facial geometry dataset ([Figure 5.1\(c\)](#)), an interesting topological challenge emerges. Due to the limited number of initial subclusters, the outer ring is partitioned into elongated curved arcs. Given their spatial extent, certain arcs obscure the underlying density gaps, inadvertently forming artificial structural bridges between the outer perimeter and the internal features. Consequently, during UniForCE’s local unimodality phase, the standard dip-test fails to perceive these density gaps. This limitation ultimately results in the incorrect agglomeration of the ring with the internal ‘left eye’ and ‘mouth’ structures.

Conversely, in the modified UniForCE framework, the application of the mud-pod test successfully detects this hidden internal multimodality. With just a single subsequent split, the algorithm effectively dismantles the artificial bridge, allowing for the correct topological separation of the distinct structures. Ultimately, this intervention ensures the precise identification of all ground-truth clusters, yielding a remarkably high AMI score.

Next, we examine the Twodiamonds dataset ([Figure 5.1\(d\)](#)) which features two diamonds next to each other. Notably, during the overclustering step, the algorithm generated a specific subcluster positioned exactly at the intersection connecting the two diamond-shaped structures. The baseline UniForCE falsely assumed that all initial subclusters, including this intersectional one, were strictly homogeneous. Consequently, during the subcluster aggregation phase, this specific subcluster acted as artificial structural ‘bridge’. By satisfying the unimodal pair tests with adjacent subclusters in both the left and right rhombuses, it propagated a false connection, causing the algorithm to merge the two distinct shapes into a single final cluster.

In contrast, SMCUT framework intervenes immediately after the generation of the initial 7 subclusters. By leveraging the Mahalanobis distances over linear random projections, the mud-pod algorithm successfully identified the hidden density gap within that critical intersectional subcluster, recognizing it as multimodal. This early detection prompted an immediate split, resulting in truly unimodal and homogeneous subclusters. Subsequently, during the construction of the unimodality graph, the underlying density gap between the two rhombuses was completely exposed. Without this artificial bridge, the unimodal pair testing could no longer find a path between the rhombuses resulting in two distinct shapes.

From this behavior, it can be concluded that the integration of the mud-pod test significantly enhances the algorithm’s robustness, allowing it to recover from incorrect or misleading initial subclustering generated in its respective phase.

Furthermore, in the case of the Complex2D dataset ([Figure 5.1\(e\)](#)), the limitations of the baseline method become apparent. Because the initial overclustering resolution is relatively low, the initial phase generates spatially extended subclusters. However, the mud-pod criterion immediately recognizes the severe multimodality within these massive structures and partitions them into smaller and statistically homogeneous components.

On the contrary, since the baseline algorithm lacks an internal splitting mechanism following the overclustering step, it is trapped with these incorrect and structurally flawed subclusters. As a result, during the unimodal pair test phase, the standard dip-test rejects almost every case, as the pairwise union of subclusters which are already multimodal continuously fail the unimodality tests, halting the graph construction prematurely.

In conclusion, once again we can see how the integration of the mud-pod multivariate test effectively compensates for an insufficient initial overclustering. By dynamically reconstructing the space into valid and truly unimodal blocks, the proposed algorithm can execute highly accurate topological aggregations, demonstrating remarkable resilience against poor initialization.

[Figure 5.1\(f\)](#) provides an example of a 3D dataset. As mentioned previously, the baseline algorithm assumes that the data is partitioned into homogeneous subclusters residing within convex regions of the space. However, when applied to datasets characterized by highly non-linear structures, it often fails to capture the underlying topological complexity. Therefore, it generates mixed subclusters that encompass segments from distinct ground-truth clusters and violating the assumption of local homogeneity. This violation results in the algorithms underperformance during the unimodal pair testing.

The integration of the mud-pod test effectively resolves this limitation. The modified algorithm successfully detects subclusters exhibiting multimodality and splits them into true homogeneous clusters. Consequently, these refined blocks enable the unimodality graph to discover valid unimodal paths and perform more accurate topological aggregations. This process allows SMCUT to achieve a better cluster estimation while also achieving a higher AMI score.

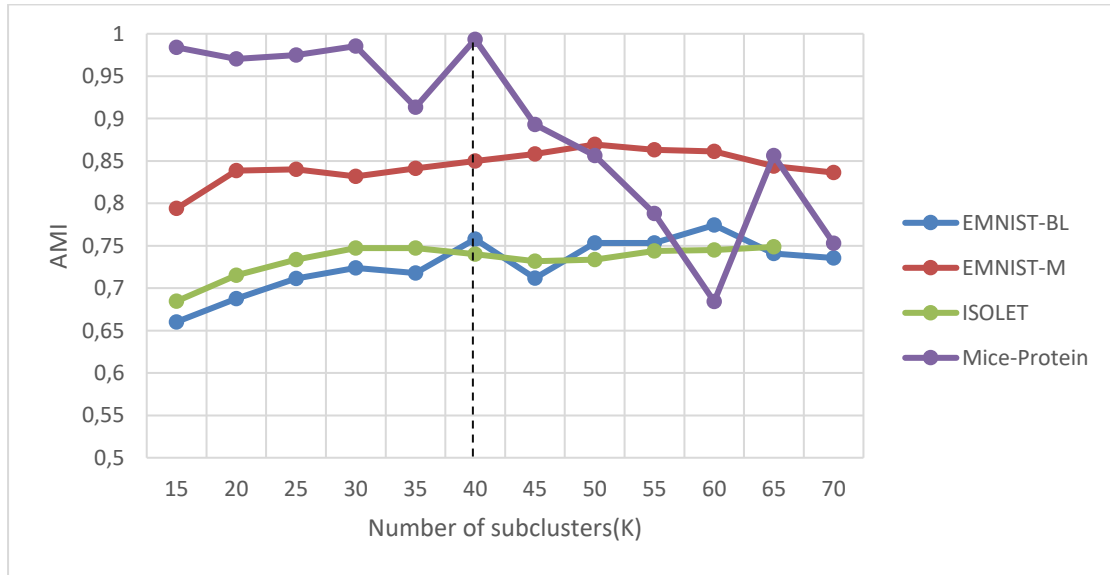
Lastly, in [Figure 5.1\(g\)](#) we have again the case of the Complex2D dataset but with 16 initial subclusters. When employing the baseline UniForCE algorithm, a notable deficiency in fully reconstructing the ring topology is evident. The conventional dip-test criterion struggles to verify local unimodality, effectively trapping the algorithm within the artificial boundaries introduced by the initial overclustering. In contrast, the

proposed method successfully identifies multimodality in specific initial subclusters and proceeds to split them. This corrective mechanism facilitates the seamless agglomeration of the entire ring perimeter while properly separate the compact clusters situated in the core.

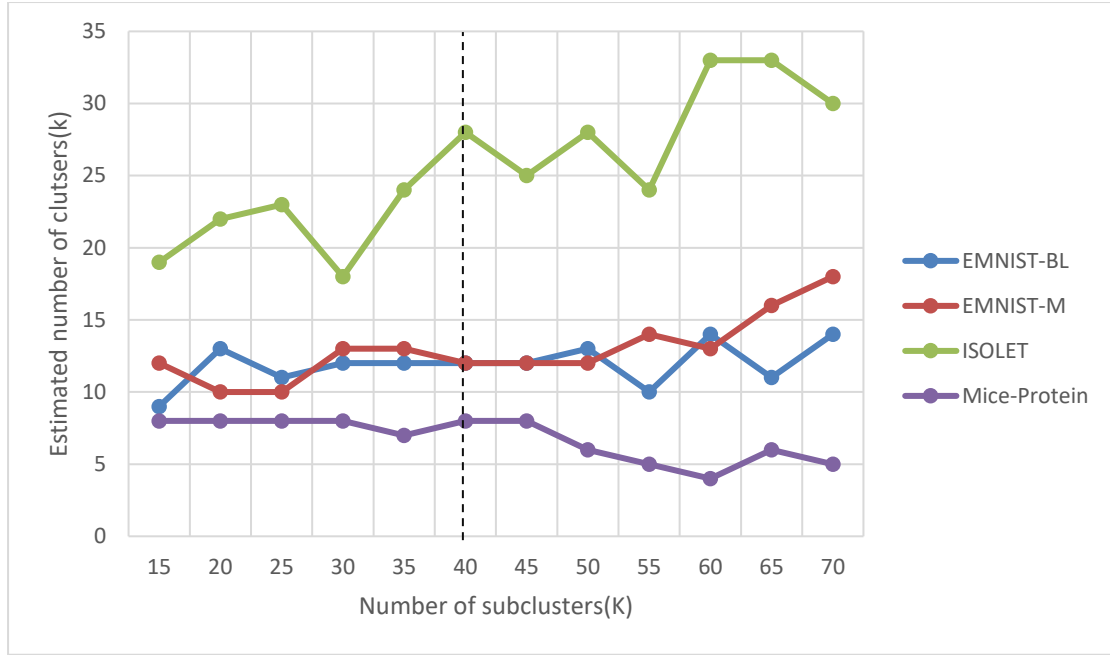
5.5 Optimal Number of Initial Subclusters (K)

The number of initial subclusters, K , is an important hyperparameter of the UniForCE framework, as it determines the initial overclustering partition of the data. A sufficiently large value of K ensures that the true underlying clusters are oversegmented into small and homogeneous subclusters that reside in convex regions. In the original UniForCE implementation a fixed value of $K=50$ was recommended, showing robustness across different datasets.

To identify the optimal value for the proposed SMCUT algorithm, a similar experiment was conducted by incrementing the value of K in steps of 5, ranging from 15 to 70. This analysis was conducted on four representative datasets: EMNIST-BL, EMNIST-M, ISOLET and Mice-Protein, covering various data distributions and dimensionalities.



(a) k vs K : Estimation of the number of clusters (k) by the SMCUT method as a function of the number of initial subclusters (K)



(b) AMI vs k: Clustering performance (AMI score) by the SMCUT method as a function of the number of initial subclusters (K).

Figure 5.2: Comparison of clustering results.

[Figure 5.2](#) summarizes the results of the hyperparameter experiment, reporting both the Adjusted Mutual Information (AMI) in [Figure 5.2\(a\)](#) and the estimated number of clusters k in [Figure 5.2\(b\)](#) as a function of K . The data reveals that for most of the tested datasets, the optimal behavior of SMCUT is observed at $K=40$.

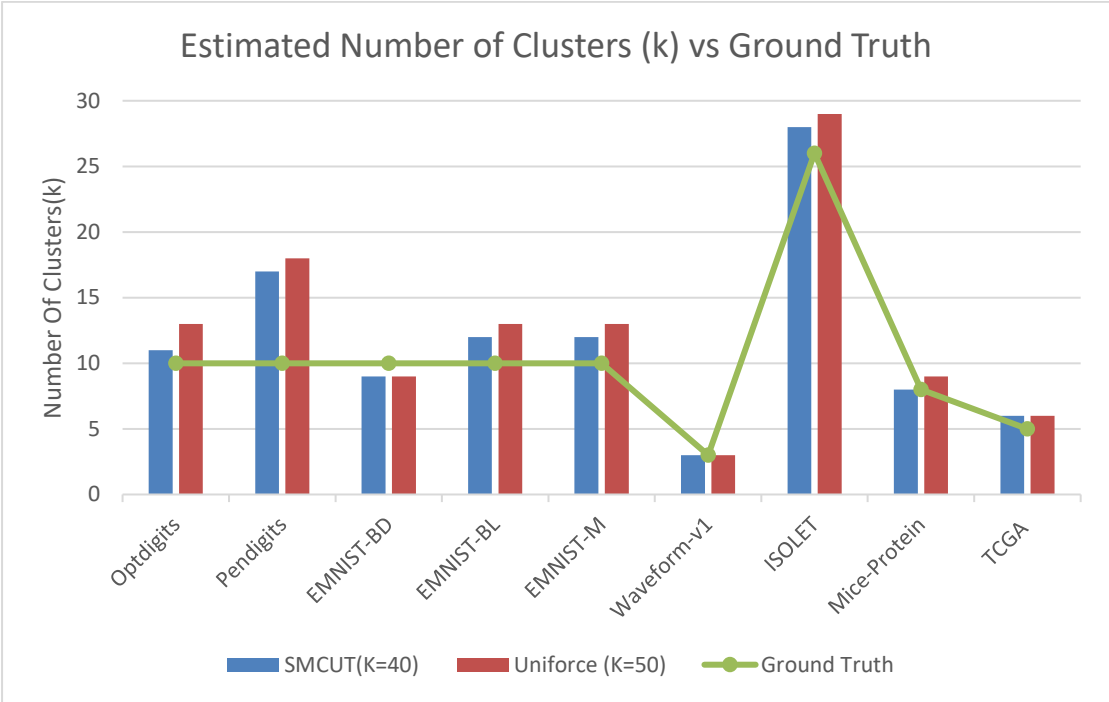
Interestingly, while the baseline UniForCE method requires $K=50$ to ensure subcluster homogeneity, our proposed method achieves equivalent or in some cases even superior results at a lower resolution. This is credited to the integration of the mud-pod test which is capable of early detecting multimodality and forcing further splits in order to reconstruct the feature space and create homogeneous subclusters.

5.6 Comparative Performance at Respective Optimal Resolutions

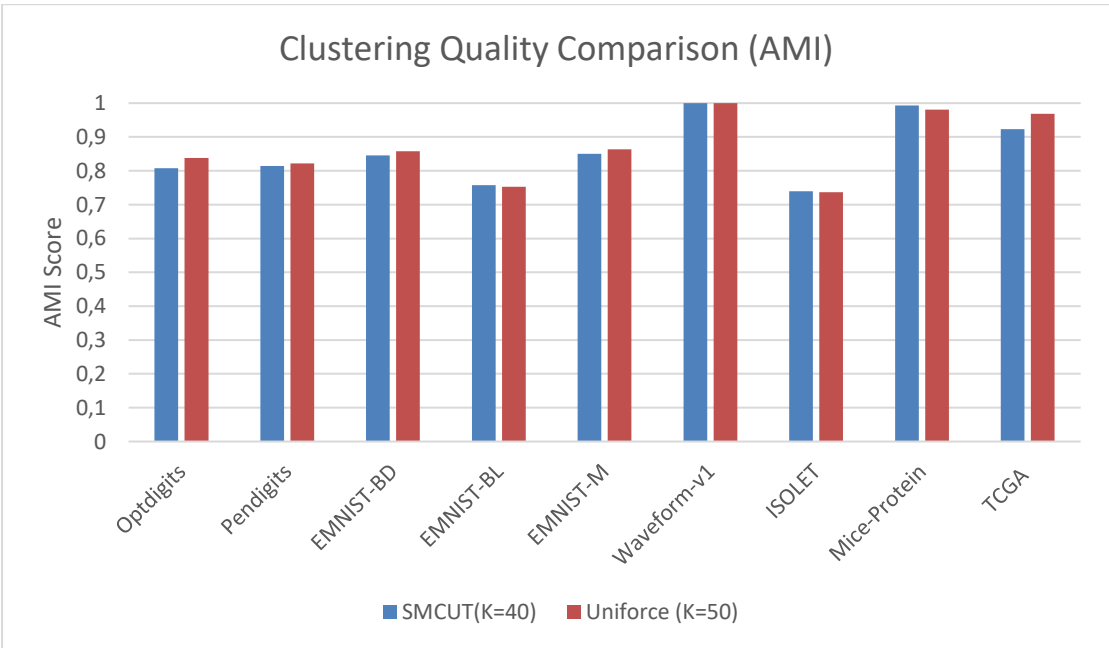
To ensure a fair and comprehensive comparison and strengthen the study even more, both algorithms were evaluated using their respective optimal hyperparameter configurations. Based on the analysis conducted in section 5.5, SMCUT was evaluated at $K=40$, while UniForCE was set to its recommended default of $K=50$.

[Figure 5.3](#) illustrates the performance across all the real datasets used, focusing once again on the Adjusted Mutual Information (AMI) and the accuracy of the estimated

number of clusters (k). A key takeaway from this comparison is that SMCUT consistently matches or outperforms the baseline algorithm in both metrics, despite utilizing a lower overclustering resolution.



(a): Estimated number of clusters for each dataset



(b): AMI score for each dataset

Figure 5.3: Comparison Experiment of SMCUT and UniForCE across four datasets

To get into more detail, by observing the datasets Optdigits, Pendigits, EMNIST-BL and EMNIST-M, we can clearly see that the proposed method demonstrates an advantage in identifying the underlying cluster structure. The estimated number of clusters by SMCUT converges significantly closer to the Ground Truth ($k=10$) compared to the original UniForCE. Regarding clustering quality, a very slight drop in AMI is observed. For EMNIST-BD, both the estimated k and the AMI score remain almost identical between the two methods.

In the waveform-v1 dataset, the performance of both algorithms on this dataset is exceptional. SMCUT successfully identifies the exact number of true clusters ($k=3$) while simultaneously achieving the maximum possible AMI score (1.0). This confirms that integrating the multivariate mud-pod check does not disrupt the clustering of well-formed, regular datasets, allowing it to perfectly capture the underlying structure.

For the ISOLET dataset, the proposed method achieved a more precise estimation of k (28) compared to the baseline's 29, against a ground truth of 26. Importantly, the clustering quality (AMI) is slightly higher for the SMCUT method. This indicates that our method managed to construct better quality clusters, while using a smaller initial number of subclusters.

For the TCGA dataset, both algorithms slightly overestimate the Ground Truth by a single cluster ($k = 6$ instead of 5). Here, a minor decrease in the AMI score is observed for SMCUT. The TCGA is characterized by extreme dimensionality ($d = 20531$). It is highly probable that reducing the initial subclusters to $K=40$, combined with mud-pod's random projections in such a massive vector space, led to the agglomeration of a specific structure that slightly impacted the mutual information metric. Nevertheless, the clustering quality remains exceptionally high.

Another interesting case is observed in the Mice-Protein dataset. SMCUT accurately detects the exact number of clusters ($k=8$), successfully correcting the baseline's overestimation ($k=9$). Concurrently, the proposed approach achieves a slightly higher AMI score, approaching 1.0. The adaptation of the mud-pod test in this dataset proved pivotal in filtering out noise and accurately grouping the protein expression profiles.

These results are particularly significant as it is demonstrated that the integration of the mud-pod test provides a superior mechanism for validating cluster homogeneity and effectively achieving more accurate estimation of the true number of clusters. This gain in efficiency confirms that our proposed method is not only more robust to parameter initialization but also more effective in capturing the topological structure of the data compared to the initial method.

5.7 Limitations and Failure Cases

The experimental evaluation of the proposed SMCUT algorithm highlighted a vulnerability when dealing with critically small subclusters. Through the analysis of the results (Tables [2,3,4](#)), it was identified that the method tends to misinterpret natural local density fluctuations or statistical noise as inherent structural multimodality. This instability, which ultimately leads to extreme over-segmentation, primarily occurs under two conditions: excessive initial overclustering and high-dimensional feature spaces.

The first major observation is that under conditions of excessive initial overclustering (i.e., a very large K), the mud-pod test exhibits increased sensitivity to small-scale fluctuations in data density. A representative case illustrating this vulnerability is the Optdigits dataset. When the algorithm is initialized with a high number of subclusters ($K=100$), the strictness of the mud-pod test inadvertently degrades the overall clustering quality. Specifically, it results in a reduced Adjusted Mutual Information (AMI) score of 0.6551 and severely underestimated the final cluster count to $k=5$. In contrast, under the exact same initialization, the baseline UniForCE algorithm proves more resilient to these local variations, achieving a significantly higher AMI of 0.8701 and estimating $k=8$ distinct clusters.

In such instances, the statistical test further partitions the already small subclusters, resulting in an extreme oversegmentation (e.g., reaching a total of 109 subclusters). This behavior suggests that when the initial overclustering is excessively high, the method tends to misinterpret natural local density fluctuations or statistical noise as inherent structural multimodality. Consequently, this excessive fragmentation hinders the final stage of the algorithm, complicating the accurate reconstruction of the broader feature space topology and leading to suboptimal cluster merging.

The second major observation is the algorithm's performance in high-dimensional spaces and again in combination with small cluster sizes. A prime example is Mice-Protein dataset ([Table 4](#)), which comprises biological profiles spanning 83 dimensions. In this instance, the UniForCE algorithm, initialized with $K=50$ subclusters, demonstrates exceptional performance, achieving an AMI of 0.9519. However, the application of the mud-pod test under the same initialization settings results in a noticeable performance degradation, reducing the AMI score to 0.8375. In this high-dimensional context, the statistical test inadvertently triggers excessive fragmentations, increasing the total number of subclusters and ultimately distorting the final estimated cluster count.

This phenomenon is naturally linked to the underlying mechanisms of the linear random projections employed by the mud-pod algorithm. Under conditions of excessive overclustering within high-dimensional feature spaces, the generated subclusters become critically small in population and spatially sparse. Operating under these constraints compromises the foundational statistical assumptions of the test. Consequently, this instability frequently leads to the erroneous rejection of the unimodal null hypothesis, causing the algorithm to falsely classify homogeneous regions as multimodal.

While the preceding analysis focused on the algorithm's limitations across real-world datasets, examining its performance on synthetic data provides critical insights. A prominent example of such a failure case is observed in [Figure 5.1\(g\)](#). Constrained by the low initial subcluster count and the tight geometric proximities between distinct shapes, the statistical criterion is unable to separate the outer ring from the central 'X' structure, ultimately leading to a flawed cluster merging.

Further analysis of the complex synthetic topology (see [Figure 5.1\(b\)](#)) reveals two additional distinct failure cases, highlighting the sensitivity of local unimodality assessments in the presence of tight spatial constraints or noise.

Regarding the merging of the extended curve with the smaller adjacent structure on the right, the results demonstrate the vulnerability of local unimodality when dealing with extreme geometric proximity or substantial noise. This outcome can be attributed to two potential scenarios: either the mud-pod test evaluated the initial subclusters in that specific region as unimodal – thus preventing any further fragmentation – or fragmentation did indeed occur, but the extreme proximity of the resulting subclusters obscured the density gap. Consequently, during the subsequent dip-test application, the lack of an obvious density gap led to the incorrect agglomeration of these distinct geometric curves.

Another interesting behavior is observed between the outer ring and the right inner cluster following the application of the mudpod criterion. It is highly probable that the mudpod algorithm detected underlying multimodality within the inner cluster, prompting its further division. However, this corrective action inadvertently generated excessively small subclusters. Consequently, when the algorithm proceeded to the unimodal pair testing phase, the tight proximity of the structures, compounded by the small sample size of these newly formed subclusters, rendered the existing density gap statistically insignificant. This cascading effect led to a false confirmation of local

unimodality and, ultimately, the flawed cluster merging of the outer ring with the distinct inner structure.

Chapter 6. Epilogue

6.1 Conclusion

In this thesis, we introduced a statistically validated clustering pipeline by integrating the mud-pod multivariate unimodality test into the UniForCE clustering framework. While UniForCE can effectively capture complex and arbitrary-shaped clusters by finding connected components in a unimodality graph, its foundational step relies on an initial overclustering phase that assumes the resulting subclusters are homogeneous and unimodal.

Our contribution reinforces this foundational step. By applying the mud-pod test immediately after the overclustering phase, we ensure that every subcluster is statistically proven to be unimodal.

We conducted a series of experiments to test our proposed algorithm. We experimented on both real-world and synthetic datasets. The results indicated that our approach achieved higher or comparable performance in both sets of datasets, taking into consideration the AMI score and the estimated number of clusters. We also performed experiments to find the optimal number of initial clusters (K). The results indicated that an optimal K would be $K=40$ which is interesting since it is less than the $K=50$, proposed in the baseline UniForCE algorithm. This is owed to the mud-pod test since it can detect multimodality and can enforce further splits to ensure unimodality among the subclusters. Another experiment we performed was a comparison of the performance of the two algorithms on their respective resolutions (K). The findings indicate that SMCUT consistently matched or outperformed UniForCE in both AMI and the estimated number of clusters, even though it utilizes a lower overclustering resolution.

Despite the strong theoretical foundation and empirical success, our approach has some limitations. The first one is that utilizing the mud-pod test while having an excessive initial overclustering resolution results in the mud-pod test exhibiting an increased sensitivity in small-scale fluctuations in data density. This results in an inaccurate reconstruction of the broader feature space topology and leading to suboptimal cluster merging. A similar behavior is observed in high-dimensional spaces where if we perform an extensive overclustering, the performance of the algorithm deteriorates.

6.2 Future work

In the current architecture, initial subclusters containing fewer than M data points are eliminated after the unimodality testing phase, and their points are redistributed to neighboring subclusters using the k-means assignment rule. However, this redistribution is performed heuristically, meaning that the receiving subclusters are not re-evaluated to ensure that the newly absorbed points do not disrupt their statistical unimodality. A promising future direction would be to develop a variant of the algorithm that introduces a post-redistribution validation step. This framework would systematically re-test the target subclusters after they absorb external points, thereby preventing the degradation of the cluster homogeneity.

Another future direction could be to further optimize the computational cost. This could be achieved by investigating the use of GPU-accelerated parallelization.

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